Data Ethics: Understanding Big Data and Algorithmic Bias

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Digital Feminisms
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Northeastern University
NULab for Digital Humanities and Computational Social Science

Feel free to ask questions at any point during the presentation!

Workshop Agenda:

- Introduction to big data
- The personal impact of big data
- Big data and privacy
- Big data, representation, and bias
- Using data responsibly
- Resources for further exploration

Slides and materials available at: bit.ly/sp26-savit-dataethics

Questions to consider:

- Who is represented online? How are we being represented online?
- How does big data impact people?
- What does this mean for conducting research with Internet data?

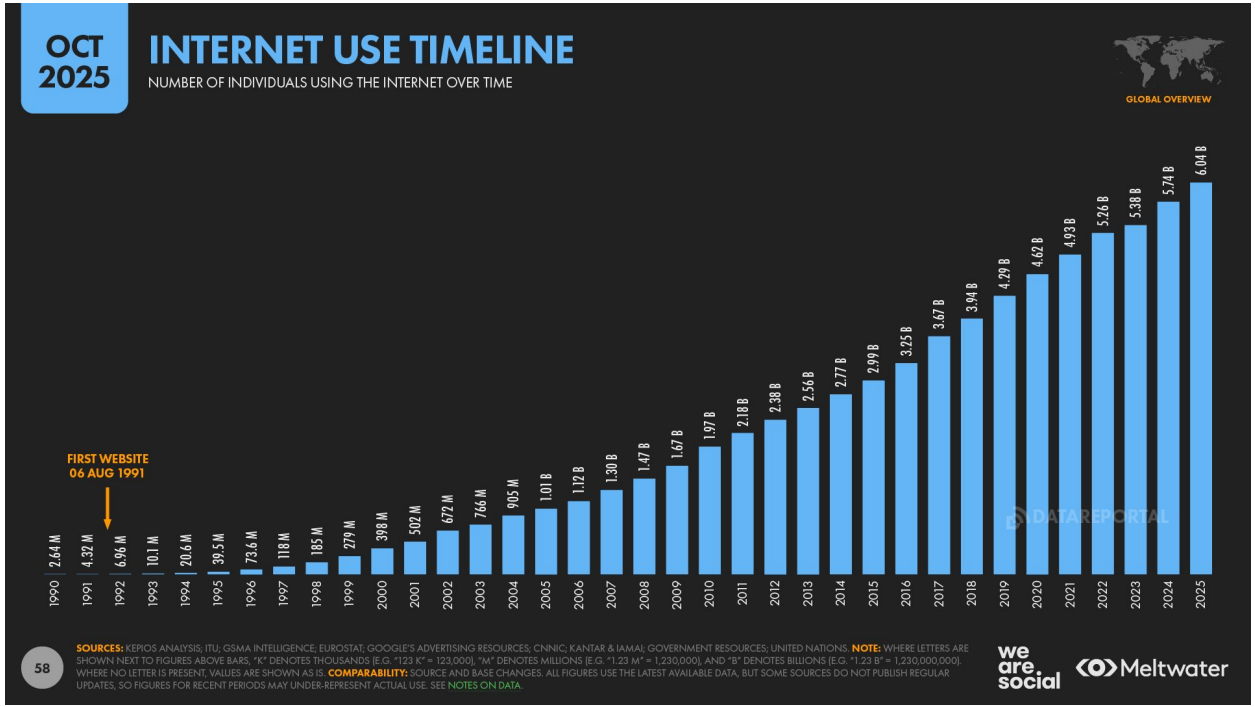
What is “Big Data”?

Feel free to ask questions at any point during the presentation!

Defining Big Data

- "Big data refers to extremely large and diverse collections of structured, unstructured, and semi-structured data that continues to grow exponentially over time" (Google Cloud).
- Big data is often collected through digital technologies and repurposed for profit (Lipartito, 2025, Surveillance Capitalism: Origins, History, Consequences, *Histories*).
- This can harm individual privacy (Wikipedia) and agency (Joseph, 2025, The algorithmic self: how AI is reshaping human identity, introspection, and agency, Frontiers in Psychology).

Big Data is Getting Bigger: Internet Use



- Internet usage is constantly increasing
- This increases the amount of data available online

Source: we are social, Meltwater, and DataReportal

Internet Use Geographically (1/2)

OCT 2025

WORLD'S LARGEST CONNECTED POPULATIONS

COUNTRIES WITH THE GREATEST NUMBER OF PEOPLE USING THE INTERNET

GLOBAL OVERVIEW

#	COUNTRY	USERS	vs. POP	▲YOY
01	CHINA	1,296,394,000	91.6%	-0.1%
02	INDIA	1,026,954,000	70.0%	+27.7%
03	UNITED STATES OF AMERICA	323,888,000	93.1%	+0.5%
04	INDONESIA	230,448,000	80.5%	+8.7%
05	BRAZIL	184,997,000	86.9%	+1.1%
06	RUSSIAN FEDERATION	135,676,000	94.4%	-0.6%
07	PAKISTAN	116,839,000	45.6%	+1.3%
08	MEXICO	110,345,000	83.5%	+1.0%
09	NIGERIA	108,700,000	45.5%	+2.3%
10	JAPAN	106,933,000	87.0%	-0.5%

#	COUNTRY	USERS	vs. POP	▲YOY
11	EGYPT	98,211,000	82.7%	+2.4%
12	PHILIPPINES	98,025,000	83.8%	+0.8%
13	VIETNAM	85,621,000	84.2%	+0.6%
14	BANGLADESH	82,806,000	47.0%	+6.9%
15	GERMANY	78,454,000	93.5%	-0.7%
16	TURKEY	77,466,000	88.3%	+0.2%
17	IRAN	73,751,000	79.6%	+0.9%
18	UNITED KINGDOM	68,090,000	97.8%	+0.6%
19	THAILAND	67,826,000	94.7%	+3.7%
20	FRANCE	63,449,000	95.2%	+0.2%

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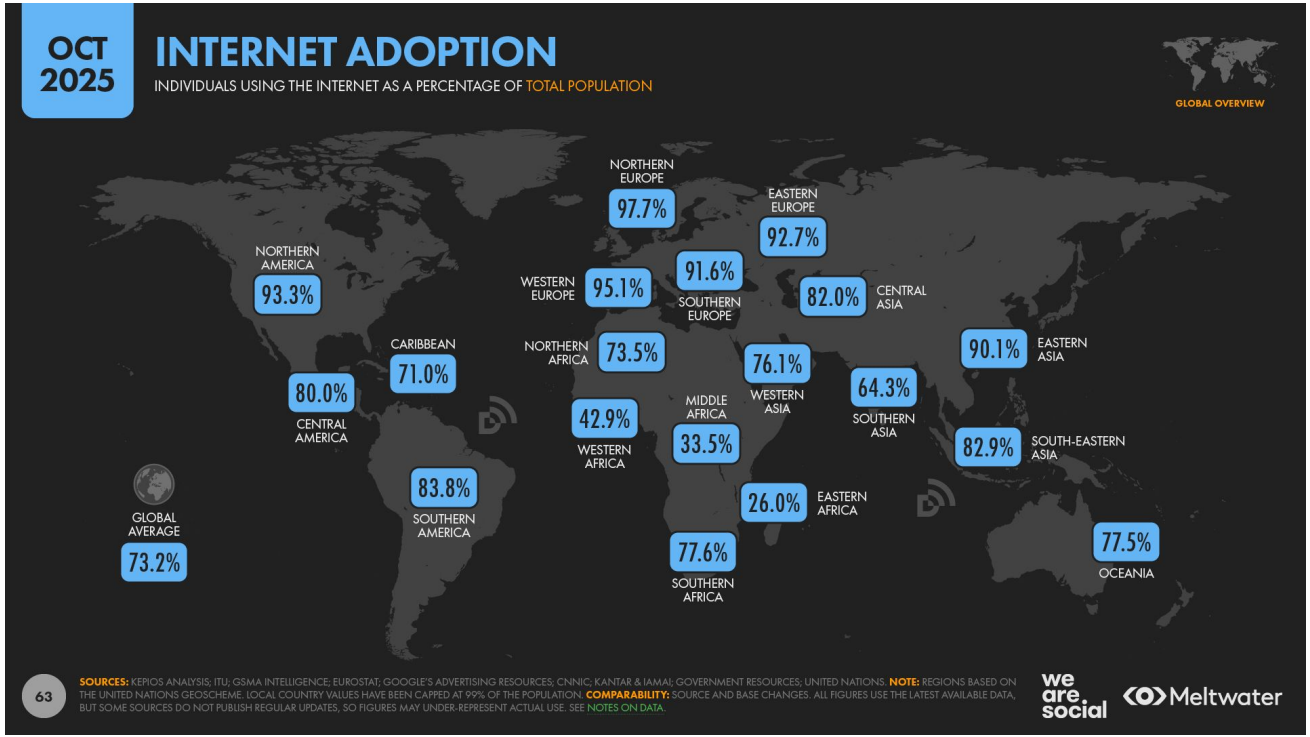
SOURCES: KEPIOS ANALYSIS; ITU; GSMA INTELLIGENCE; EUROSTAT; GOOGLE'S ADVERTISING RESOURCES; CNNIC; KANTAR & IAMAI; LOCAL GOVERNMENT AUTHORITIES; UNITED NATIONS. **NOTE:** VALUES HAVE BEEN CAPPED AT 99% OF THE POPULATION. **COMPARABILITY:** SOURCE AND BASE CHANGES. ALL FIGURES USE THE LATEST AVAILABLE DATA, BUT SOME SOURCES DO NOT PUBLISH REGULAR UPDATES, SO FIGURES MAY UNDER-REPRESENT ACTUAL USE. SEE NOTES ON DATA.

we are social Meltwater

- Internet use is distributed unevenly across countries

Source: we are social, Meltwater, and DataReportal

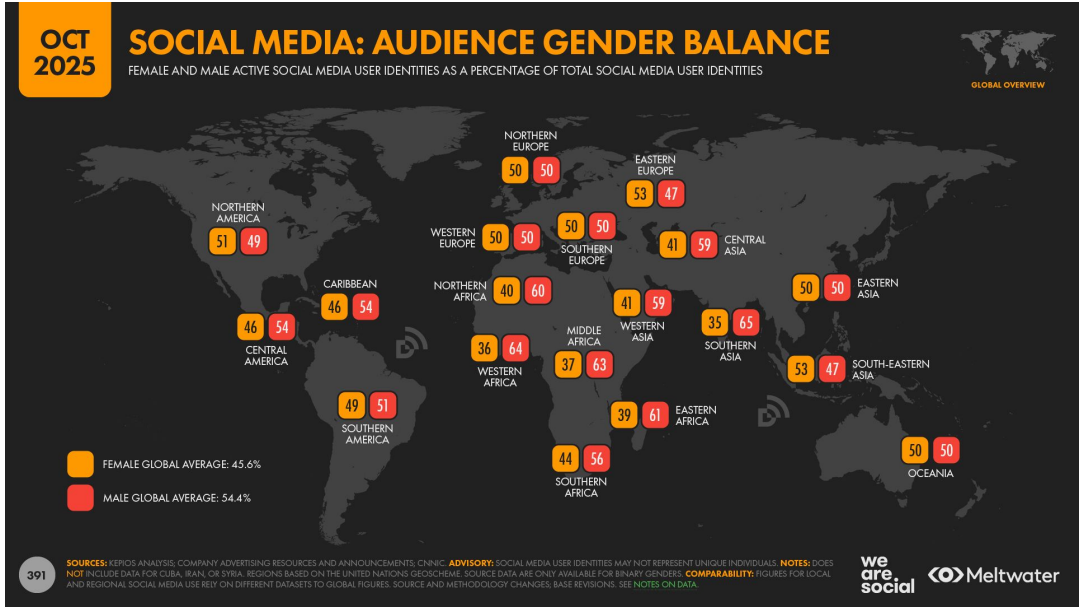
Internet Use Geographically (2/2)



- Internet use is distributed unevenly across continents

Source: we are social, Meltwater, and DataReportal

Social Media Use by Gender



Source: we are social, Meltwater, and DataReportal

- According to DataReportal, "men are roughly 7 percent more likely to use the internet than women, and there are still nearly 240 million more men using the internet than there are women online" as of 2025.
- This is reflected on social media as well

Feel free to ask questions at any point during the presentation!

Discussion: Representation

- Given trends in Internet and social media use, who is likely to be represented in and who is likely to be excluded from big data?
- How might global big data be biased?
- How might disparities in technical access impact an image of the world that is informed by big data?

Feel free to ask questions at any point during the presentation!

Why should we care about Big Data? (1/2)

- Big data sources include: digitized records, internet activity, and sensors from the physical environment.
- Big data is often privately owned and oversight is limited.
- The scale of big data enables those who use, develop, and control it to magnify their influence.
- Some websites monetize data in a "data exploitation market," selling their users' personal information.
- Big data can be used to (inadvertently or purposefully) entrench stereotypes or reproduce results that harm individuals and communities.

Source: A Matter of Scale by Matthew Jockers and Julia Flanders

Why should we care about Big Data? (2/2)

- Big data can provide researchers with unprecedented access to records
- The scale of big data enables more "exemplary, representative" theories based on larger sample sizes
- Big data is not fundamentally careless or of low quality, but one strategy or tool that you should employe with intentionality
- For questions that don't require high levels of detail or precision, big data can be helpful: "Big data isn't perfect, so let's use it in cases where perfection doesn't matter"
- Big data can connect the micro to the macro

Source: A Matter of Scale by Matthew Jockers and Julia Flanders

Technology and Data are not Neutral

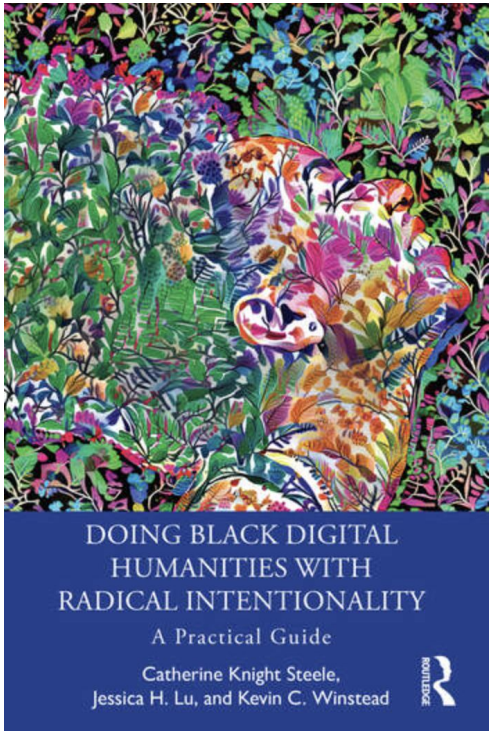
- Information systems, data collection and analysis platforms, and algorithms for working with big data can reinforce systemic, political, and cultural biases.
- These technologies are affected by input data, the way that data is presented, how the data is interpreted, and more.
- This means we also have the ability to challenge the biases, norms, and forms of discrimination embedded in technology

[The Library of Missing Datasets](#) (series), 2016 - ongoing by [Mimi Onuoha](#)

Feel free to ask questions at any point during the presentation!

Discussion: *Doing Black Digital Humanities with Radical Intentionality* by Catherine Knight Steele et al.

- How might our personal experiences shape our perception of data? (p. 9)
- How can we prioritize and center the care of people when using big data? (p. 3)
- How might we maintain personal accountability for our use of big data? (p. 8)



Feel free to ask questions at any point during the presentation!

Personal Impact of Big Data

Feel free to ask questions at any point during the presentation!

How do we contribute to Big Data?

- Entertainment and social media
- Healthcare and medical services
- Shopping and marketing
- News and information
- Travel and transportation
- Education and employment
- Public policy and safety
- How else?

Personal information can be collected by digital platforms and tools, causing data privacy issues as well as concerns of using personal data for company profit.

Terms & Conditions: Google

Categories of information we collect

Identifiers and similar information such as your name and password, phone number, and address, as well as unique identifiers tied to the browser, application, or device you're using. Some Google services like YouTube Studio provide the option to submit a valid ID (such as a passport or driver's license) to verify your identity to use additional features.

Demographic information, such as your age, gender and language. If you choose to use optional features like YouTube Creator Demographics, you can also provide additional information, like your gender identity or race and ethnicity.

Commercial information such as your payment information and a history of purchases you make on Google's services.

Biometric information if you choose to provide it, such as fingerprints in Google's product development studies.

Internet, network, and other activity information such as your search terms; views and interactions with content and ads; Chrome browsing history you've synced with your Google Account; information about the interaction of your apps, browsers, and devices with our services (like IP address, crash reports, and system activity); and activity on third-party sites and apps that use our services. You can review and control activity data stored in your Google Account in My Activity.

Geolocation data, such as may be determined by GPS, IP address, and other data from sensors on or around your device, depending in part on your device and account settings. Depending on these settings, this may include **precise location data**, for example GPS data for Android features like navigation or finding your phone. Learn more about Google's use of location information.

Audio, electronic, visual, and similar information, such as voice and audio information.

Communications data, such as emails, if you use our services to send and receive messages.

Health information if you choose to provide it, such as your medical history, vital signs and health metrics (like blood glucose levels), and other similar information related to your physical or mental health, in the course of using Google services that offer health-related features, such as the Google Health Studies app, Fitbit, or Pixel devices..

Google's Privacy Policy:
https://policies.google.com/?hl=en-US

Terms & Conditions: Boston Digital Services

CITY of BOSTON | Mayor Michelle Wu

INFORMATION AND SERVICES PUBLIC NOTICES

TERMS OF USE PRIVACY POLICY POLICIES FOR SPECIFIC SERVICES

INFORMATION WE AUTOMATICALLY COLLECT AND RECEIVE

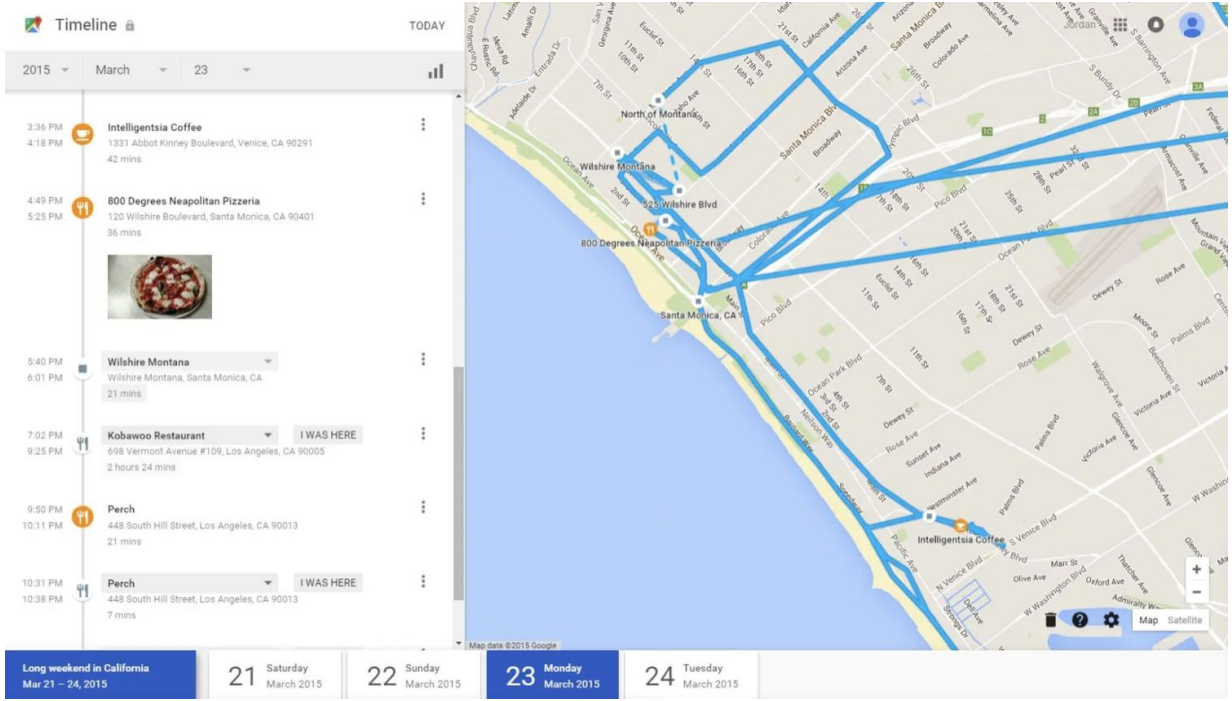
Some information is automatically collected when you visit Boston.gov or any digital City service. Like many other websites, w cookies, metadata, and geocodes. The information automatically collected may include, but is not limited to:

- the domain name of your Internet Service Provider and/or your computer
- the IP address from which you access our website
- your browser type (e.g., Google Chrome, Internet Explorer, or Mozilla Firefox)
- your operating system (e.g., Windows or Mac)
- the type of device you are using
- the date and time you accessed our Services
- location data

What information are you comfortable with being collected and shared by companies, governments, and third-party services?

City of Boston Digital Services Privacy Policy

Example: Google Maps' Timeline



If you have **location services** turned on for Google (for instance, if you use Google maps), Google can track your phone's location. If you use Google maps, try going to the ribbon icon that says "You" and then select "Timeline" at the bottom.

[Hands on with Google Maps' Your Timeline: Fascinating, but freaky](#) by Jordan Novet (2015)

Example: Instagram and Advertising

- In "'It ain't a compliment": Feminist data visualisation and digital street harassment advocacy' Fileborn and Trott (2022) describe how individuals' experiences of harassment when disclosed on Instagram could be vulnerable to monetization by advertisers

Instagram posts by the account Ourstreetsnow, found in Fileborn and Trott (2022)

Example: Flo Ovulation & Period Tracker

- In 2021, the Federal Trade Commission alleged that "Flo disclosed health data from millions of users of its Flo Period & Ovulation Tracker app to third parties that provided marketing and analytics services to the app, including Facebook's analytics division, Google's analytics division, Google's Fabric service, AppsFlyer, and Flurry."
- This information included a user's pregnancy

FTC Health App Guidance

Discussion: Datafication

- In what ways does “datafication” impact your day-to-day life?
- Have you seen instances of apps customizing what is shown to you based on data collected about you?
- How do you feel this impacts your decision making?

Feel free to ask questions at any point during the presentation!

Big Data & Privacy

Feel free to ask questions at any point during the presentation!

Data Privacy

- It's important to pay attention to data privacy when using digital resources
- Data privacy is a person's ability to control what of their personal information is shared and with whom.
- To help you make informed decisions about interacting with digital tools in ways that honor your boundaries with your data and/or personal information, the DITI has prepared a handout on Data Privacy

Downloading Your Data

- **Meta** (Facebook & Instagram): Go to Accounts Center (https://accountscenter.facebook.com/) > Your information and permissions > Export your information
- **Google**: How to download your Google data (https://support.google.com/accounts/answer/3024190?hl=en)

Maintaining Privacy: Online Activity

- There are ways to identify what information websites collect about their users.
- Be sure to access the Terms and Conditions offered by a website to see what the site is disclosing about how they track you.
- Blacklight is a "real-time website privacy inspector" developed by *The Markup*, a nonprofit publication that investigates data misconduct. You can use it to scan and reveal the specific user-tracking technologies on any site.

Maintaining Privacy: Offline Activity

- Consider where you bring your phone as its connection to nearby cell towers and other devices can be used to record your location.
- Encrypt the data on your phone
- Consider using encrypted messaging apps and disabling biometric unlock features
- If you do bring your phone to an event, consider placing it in airplane mode and not unlocking it when taking pictures
- Note that photos and videos uploaded to social media may include location information and time stamps
    - To help avoid this, turn off your phone's location services

Discussion: Data Collection

- Who holds the power over online and technical data collection?
- How is the data collected?
- Whose privacy is impacted by its collection?

Moving Forward

How can we use data responsibly?

Feel free to ask questions at any point during the presentation!

Alleviating Injustice: Development

- When we look at the data used to train an algorithm, we must ask what kinds of data are being counted, and what kinds of data are being overlooked, ignored, and excluded?

> **"Algorithms by themselves are neither good nor bad. It is merely a question of taking care in how they are built."**
> - Sendhil Mullainathan, Chicago Booth University

Alleviating Injustice: Counting

- What are the consequences of counting and not counting different kinds of data on various populations, especially marginalized groups?
- Ask: will the data-driven solution reduce human bias or amplify it?

"Counting and measuring do not always have to be tools of oppression. We can also use them to hold power accountable, to reclaim overlooked histories, and to build collectivity and solidarity." Catherine D'Ignazio & Lauren Klein, *Data Feminism*, 2020

Activity: Group Question Creation

- Take the next few minutes to discuss with those around you and jot down some questions you would ask yourself or others before conducting data-related research.
  - For example: What information is being collected and from where? To whom does this data belong?
- Add one or two of these questions to the shared Google Doc, and then we'll go through each as a class.
- Google Document Link: https://bit.ly/digital-feminisms-document

Questions Before Starting Research

- What information is being collected and from where? What is its context? To whom does this data belong?
- How is it being collected? Do participants know that it is collected, how it will be collected, and how will it be used?
- How will the data be analyzed? What biases and ideologies may be implicit in this analysis?
- Who will this research impact? Who will it benefit? Who will it potentially harm? Have these harms been considered in collaboration with impacted communities?
- Is a technical approach appropriate for the topic?

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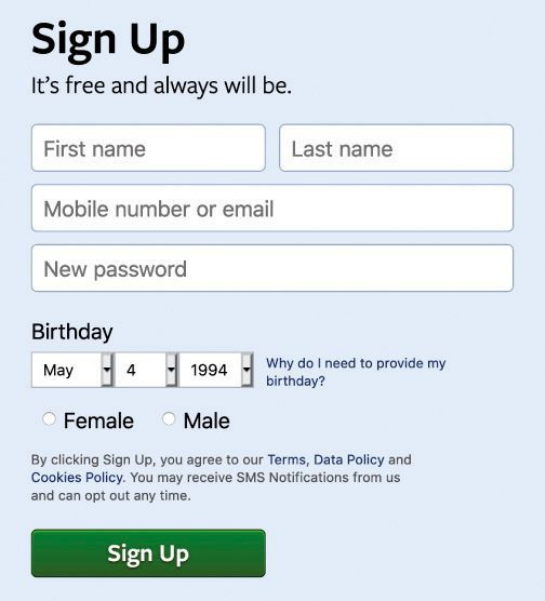
Collecting Data

Which of these is likely to collect more accurate and representative data about users' genders?



Source: Positive Voices survey from Public Health England, published in D'Ignazio & Klein, *Data Feminism*, 2020

Source: Facebook's account creation page circa 2018, published in D'Ignazio & Klein, *Data Feminism*, 2020

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Responsibly Using Data

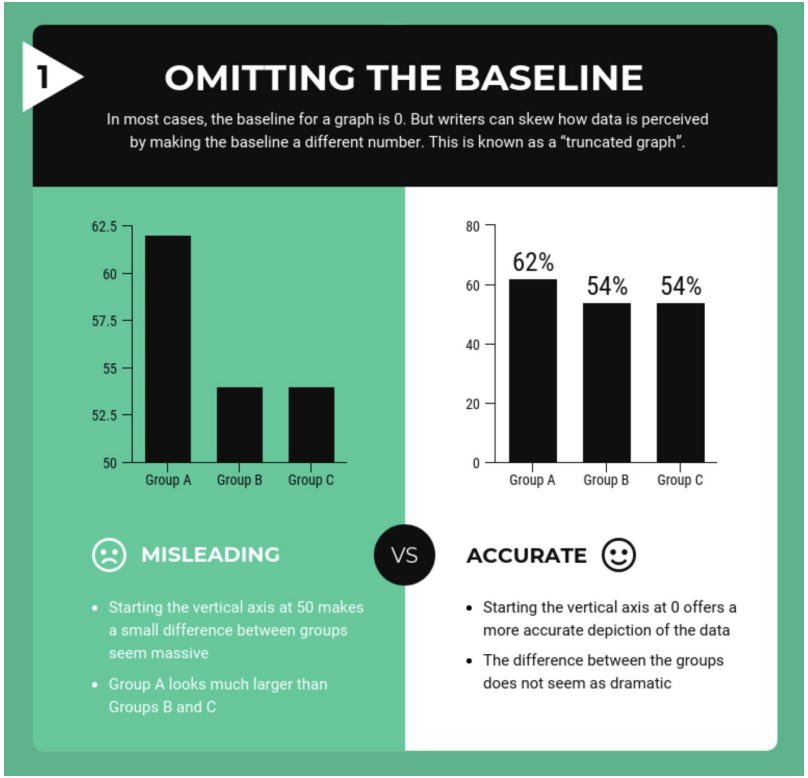
- Be data-literate—turn a critical eye to studies that use big data, evaluate the sources of that data, and carefully examine the conclusions authors draw from their sources.
- Be thoughtful and radically intentional as you incorporate big data or conclusions drawn from big data sources in your work. Think:
  - Am I fostering an "ethic of love?" (bell hooks 2006).
  - Am I pushing against the status quo? (Steele et al. 2023)
  - Am I centering BIPOC feminist traditions? (Steele et al. 2023)
  - Are there alternative sources of knowledge I can learn from? (Steele et al. 2023)

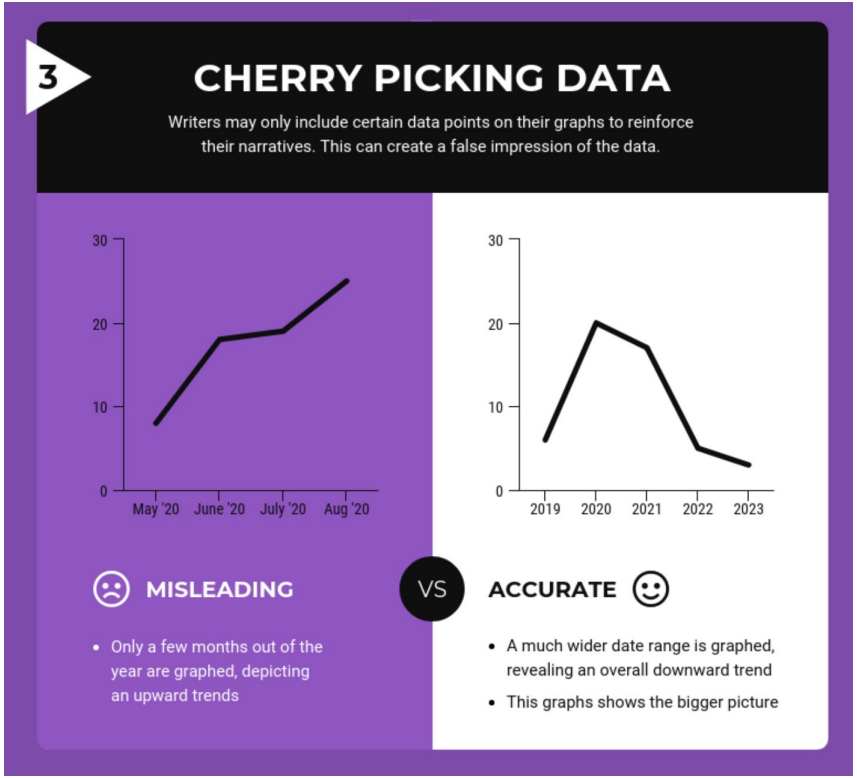
Vetting and Citing Data Sources

Regardless of the type of source you're using, you should always:

1) Try to verify the information presented in the source by finding other (independent) sources that support it.
2) Be clear in your writing about what kind of source it is, where you found it, and how you're using it (be explicit about your process and the source's purpose).
3) Cite your source appropriately so that any reader can find it.

Be Mindful of Infographics and Data Visualizations





For Further Exploration

Feel free to ask questions at any point during the presentation!

Research Networks and Projects

- Feminist Internet Research Network
- The <A+> Alliance for Inclusive Algorithms
- Algorithmic Justice League
- The Design Justice Network
- A Human Rights-Based Approach to Responsible AI by Vinodkumar Prabhakaran, Margaret Mitchell, Timnit Gebru, Iason Gabriel
- Race After Technology: Abolitionist Tools for the New Jim Code by Ruha Benjamin
- Unmasking AI by Joy Buolamwini
- Automating Inequality by Virginia Eubanks

NULab Faculty Research on Data, Algorithms, & AI

- "John Wihbey and Christo Wilson on Tiktok Data Espionage Concerns"
- "Alan Mislove Co-Authors Research on Discriminatory Ad Algorithms"
- "John Wihbey Weighs In On AI's Potential to Impact the 2024 Presidential Election"
- "Tina Eliassi-Rad Co-Creates New AI Model that Predicts Human Lifespan"
- "Nabeel Gillani Interviewed by Tech Talk Podcast on AI and Education"
- "John Wihbey Participates in a Panel on Content Moderation"
- "John Wihbey Comments on Google's New 'AI Overview'"
- "John Wihbey on the Politics of AI"
- "John Wihbey Interviewed on AI and Epistemic Risk"
- "Malik Haddad on the Regulation of AI"

Data Science and Digital Humanities at Northeastern

- Women Writers Project
- Digital Scholarship Group
- Early Caribbean Digital Archive
- DATA Club
- The Data Science Hub
- NULab for Digital Humanities and Computational Social Science
- Have a project of your own? For grants or research projects, check out the NULab's Grant Page.

DITI Resources

- Copyright and Fair Use Handout
- Data Ethics Handout
- Data Privacy Handout

Thank you!

—**Developed by:** Zhen Guo, Sara Morrell, Claire Lavarreda, Shannon Peifer, Claire Tratnyek, Vaishali Kushwaha, Yana Mommadova, Colleen Nugent, Tieanna Graphenreed, Javier Rosario, Ana Abraham & Chris McNulty

- For more information on DITI, please see: https://bit.ly/diti-about
- Schedule an appointment with us! https://bit.ly/diti-meeting
- If you have any questions, contact us at: nulab.info@gmail.com

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